```
[3]: import pandas as pd
df = pd.read_csv("/Users/miruthi/Downloads/10prog_4laptops.csv")
print(df)
pd.get_dummies(df)

from sklearn.preprocessing import OneHotEncoder
enc = OneHotEncoder()
enc_data = enc.fit_transform(df[['Category']])
print(enc_data)
print(enc.categories_ )
enc_data.toarray()
enc_df = pd.DataFrame(enc_data.toarray())
print(enc_df)
enc_df = pd.DataFrame(enc_data.toarray(), columns = ['2 in 1 Convertible',
'Gaming', 'Netbook', 'Notebook', 'Ultrabook', 'Workstation'])
print(enc_df)
df1 = df.join(enc_df)
print(df1)
```

```
    Manufacturer       Model_Name   Category  Screen_size_inches  \
0          Apple      MacBook Pro  Ultrabook                13.3
1          Apple      Macbook Air  Ultrabook                13.3
2             HP           250 G6   Notebook                15.6
3          Apple      MacBook Pro  Ultrabook                15.4
4          Apple      MacBook Pro  Ultrabook                13.3
..           ...              ...        ...                 ...
972         Dell     Alienware 17     Gaming                17.3
973      Toshiba  Tecra A40-C-1DF   Notebook                14.0
974         Asus        Rog Strix     Gaming                17.3
975           HP      Probook 450   Notebook                15.6
976       Lenovo    ThinkPad T460   Notebook                14.0
```

```
                    Storage                         GPU Operating_System  \
0                128GB SSD  Intel Iris Plus Graphics 640            macOS
1     128GB Flash Storage        Intel HD Graphics 6000            macOS
2                256GB SSD         Intel HD Graphics 620            No OS
3                512GB SSD            AMD Radeon Pro 455            macOS
4                256GB SSD  Intel Iris Plus Graphics 650            macOS
..                     ...                           ...              ...
972  256GB SSD +  1TB HDD       Nvidia GeForce GTX 1070          Windows
973              256GB SSD         Intel HD Graphics 520          Windows
974  256GB SSD +  1TB HDD       Nvidia GeForce GTX 1060          Windows
975  128GB SSD +  1TB HDD          Nvidia GeForce 930MX          Windows
976           508GB Hybrid         Intel HD Graphics 520          Windows

    Operating_System_Version  Weight_kg        Price
0                        NaN       1.37  11912523.48
1                        NaN       1.34   7993374.48
2                        NaN       1.86   5112900.00
3                        NaN       1.83  22563005.40
4                        NaN       1.37  16037611.20
..                       ...        ...          ...
972                       10       4.42  24897600.00
973                       10       1.95  10492560.00
974                       10       2.73  18227710.80
975                       10       2.04   8705268.00
976                        7       1.70   8909784.00

[977 rows x 13 columns]
<Compressed Sparse Row sparse matrix of dtype 'float64'
        with 977 stored elements and shape (977, 6)>
  Coords        Values
  (0, 4)        1.0
  (1, 4)        1.0
  (2, 3)        1.0
```

```
[977 rows x 13 columns]
<Compressed Sparse Row sparse matrix of dtype 'float64'
        with 977 stored elements and shape (977, 6)>
  Coords        Values
  (0, 4)        1.0
  (1, 4)        1.0
  (2, 3)        1.0
  (3, 4)        1.0
  (4, 4)        1.0
  (5, 3)        1.0
  (6, 4)        1.0
  (7, 4)        1.0
  (8, 4)        1.0
  (9, 4)        1.0
  (10, 3)       1.0
  (11, 3)       1.0
  (12, 4)       1.0
  (13, 3)       1.0
  (14, 4)       1.0
  (15, 4)       1.0
  (16, 3)       1.0
  (17, 4)       1.0
  (18, 3)       1.0
  (19, 4)       1.0
  (20, 2)       1.0
  (21, 1)       1.0
  (22, 3)       1.0
  (23, 0)       1.0
  (24, 4)       1.0
  :     :
  (952, 3)      1.0
  (953, 0)      1.0
  (954, 4)      1.0
```

```
        'Workstation'], dtype=object)]
        0    1    2    3    4    5
0    0.0  0.0  0.0  0.0  1.0  0.0
1    0.0  0.0  0.0  0.0  1.0  0.0
2    0.0  0.0  0.0  1.0  0.0  0.0
3    0.0  0.0  0.0  0.0  1.0  0.0
4    0.0  0.0  0.0  0.0  1.0  0.0
..   ...  ...  ...  ...  ...  ...
972  0.0  1.0  0.0  0.0  0.0  0.0
973  0.0  0.0  0.0  1.0  0.0  0.0
974  0.0  1.0  0.0  0.0  0.0  0.0
975  0.0  0.0  0.0  1.0  0.0  0.0
976  0.0  0.0  0.0  1.0  0.0  0.0

[977 rows x 6 columns]
     2 in 1 Convertible  Gaming  Netbook  Notebook  Ultrabook  Workstation
0                   0.0     0.0      0.0       0.0        1.0          0.0
1                   0.0     0.0      0.0       0.0        1.0          0.0
2                   0.0     0.0      0.0       1.0        0.0          0.0
3                   0.0     0.0      0.0       0.0        1.0          0.0
4                   0.0     0.0      0.0       0.0        1.0          0.0
..                  ...     ...      ...       ...        ...          ...
972                 0.0     1.0      0.0       0.0        0.0          0.0
973                 0.0     0.0      0.0       1.0        0.0          0.0
974                 0.0     1.0      0.0       0.0        0.0          0.0
975                 0.0     0.0      0.0       1.0        0.0          0.0
976                 0.0     0.0      0.0       1.0        0.0          0.0

[977 rows x 6 columns]
    Manufacturer       Model_Name   Category  Screen_size_inches  \
0          Apple      MacBook Pro  Ultrabook                13.3
1          Apple      Macbook Air  Ultrabook                13.3
2             HP           250 G6   Notebook                15.6
3          Apple      MacBook Pro  Ultrabook                15.4
```

```
..                   ...                        ...            ...
972  256GB SSD +  1TB HDD       Nvidia GeForce GTX 1070          Windows
973             256GB SSD         Intel HD Graphics 520          Windows
974  256GB SSD +  1TB HDD       Nvidia GeForce GTX 1060          Windows
975  128GB SSD +  1TB HDD          Nvidia GeForce 930MX          Windows
976          508GB Hybrid         Intel HD Graphics 520          Windows

    Operating_System_Version  Weight_kg        Price  2 in 1 Convertible  \
0                        NaN       1.37  11912523.48                 0.0
1                        NaN       1.34   7993374.48                 0.0
2                        NaN       1.86   5112900.00                 0.0
3                        NaN       1.83  22563005.40                 0.0
4                        NaN       1.37  16037611.20                 0.0
..                       ...        ...          ...                 ...
972                       10       4.42  24897600.00                 0.0
973                       10       1.95  10492560.00                 0.0
974                       10       2.73  18227710.80                 0.0
975                       10       2.04   8705268.00                 0.0
976                        7       1.70   8909784.00                 0.0

     Gaming  Netbook  Notebook  Ultrabook  Workstation
0       0.0      0.0       0.0        1.0          0.0
1       0.0      0.0       0.0        1.0          0.0
2       0.0      0.0       1.0        0.0          0.0
3       0.0      0.0       0.0        1.0          0.0
4       0.0      0.0       0.0        1.0          0.0
..      ...      ...       ...        ...          ...
972     1.0      0.0       0.0        0.0          0.0
973     0.0      0.0       1.0        0.0          0.0
974     1.0      0.0       0.0        0.0          0.0
975     0.0      0.0       1.0        0.0          0.0
976     0.0      0.0       1.0        0.0          0.0

[977 rows x 19 columns]
```

[5]:
```python
from sklearn.preprocessing import LabelEncoder
le = LabelEncoder()

df['RAM'] = le.fit_transform(df['RAM'])
print(df)

print(le.classes_ )
```

```
    Manufacturer       Model_Name   Category  Screen_size_inches  \
0          Apple      MacBook Pro  Ultrabook                13.3
1          Apple      Macbook Air  Ultrabook                13.3
2             HP           250 G6   Notebook                15.6
3          Apple      MacBook Pro  Ultrabook                15.4
4          Apple      MacBook Pro  Ultrabook                13.3
..           ...              ...        ...                 ...
972         Dell     Alienware 17     Gaming                17.3
973      Toshiba  Tecra A40-C-1DF   Notebook                14.0
974         Asus        Rog Strix     Gaming                17.3
975           HP      Probook 450   Notebook                15.6
976       Lenovo    ThinkPad T460   Notebook                14.0

                                Screen                          CPU  RAM  \
0    IPS Panel Retina Display 2560x1600         Intel Core i5 2.3GHz    7
1                             1440x900         Intel Core i5 1.8GHz    7
2                     Full HD 1920x1080   Intel Core i5 7200U 2.5GHz    7
3    IPS Panel Retina Display 2880x1800         Intel Core i7 2.7GHz    1
4    IPS Panel Retina Display 2560x1600         Intel Core i5 3.1GHz    7
..                                  ...                          ...  ...
972                   Full HD 1920x1080  Intel Core i7 6700HQ 2.6GHz    4
973                   Full HD 1920x1080   Intel Core i5 6200U 2.3GHz    7
974                   Full HD 1920x1080  Intel Core i7 7700HQ 2.8GHz    1
975         IPS Panel Full HD 1920x1080  Intel Core i5 7200U 2.70GHz    7
976                            1366x768   Intel Core i5 6200U 2.3GHz    5
```

```
976                          1366x768   Intel Core i5 6200U 2.3GHz    5

                  Storage                          GPU Operating_System  \
0               128GB SSD  Intel Iris Plus Graphics 640            macOS
1     128GB Flash Storage        Intel HD Graphics 6000            macOS
2               256GB SSD         Intel HD Graphics 620            No OS
3               512GB SSD            AMD Radeon Pro 455            macOS
4               256GB SSD  Intel Iris Plus Graphics 650            macOS
..                    ...                            ...              ...
972  256GB SSD +  1TB HDD       Nvidia GeForce GTX 1070          Windows
973             256GB SSD         Intel HD Graphics 520          Windows
974  256GB SSD +  1TB HDD       Nvidia GeForce GTX 1060          Windows
975  128GB SSD +  1TB HDD          Nvidia GeForce 930MX          Windows
976          508GB Hybrid         Intel HD Graphics 520          Windows

    Operating_System_Version  Weight_kg        Price
0                        NaN       1.37  11912523.48
1                        NaN       1.34   7993374.48
2                        NaN       1.86   5112900.00
3                        NaN       1.83  22563005.40
4                        NaN       1.37  16037611.20
..                       ...        ...          ...
972                       10       4.42  24897600.00
973                       10       1.95  10492560.00
974                       10       2.73  18227710.80
975                       10       2.04   8705268.00
976                        7       1.70   8909784.00

[977 rows x 13 columns]
[0 1 2 3 4 5 6 7]
```

```
[ ]:
```